KAIZEN MATHS

Lesson starter

Correlation and Regression

Prediction & Interpretation

Name _____

SA1

The table shows revision time (hours) and test score (marks) for 6 observations. Find the regression line of y on x, then estimate test score when revision time is 4 hours. State whether this is interpolation or extrapolation.

Observation	x	y	x^2	xy
1	1	39	1	39
2	2	48	4	96
3	3	52	9	156
4	4	61	16	244
5	5	65	25	325
6	6	74	36	444
Total	21	339	91	1304

Show your working

KAIZEN MATHS

Lesson starter

Correlation and Regression

Prediction & Interpretation

Name _____

SB1

For advertising spend and weekly sales, the regression line is $\hat{y} = 22.000 + 9.000x$. The observed x values were between 2 and 7. Estimate weekly sales when advertising spend is 10 hundreds of pounds, and comment on reliability.

Show your working

KAIZEN MATHS

Lesson starter

Correlation and Regression

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Name _____

SC1

The table shows age of car (years) and resale value (thousands of pounds) for 6 observations. The regression line of y on x is $\hat{y} = 24.267 - 2.171x$. For the observation where x = 3, calculate the residual.

Observation	x	y
1	1	23
2	2	19
3	3	18
4	4	14
5	5	15
6	6	11

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KAIZEN MATHS

Lesson starter

Correlation and Regression

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Name _____

SD1

The table shows advertising spend (hundreds of pounds) and weekly sales (items) for 6 observations. Find the regression line of y on x, then estimate weekly sales when advertising spend is 5 hundreds of pounds. State whether this is interpolation or extrapolation.

Observation	x	y	x^2	xy
1	2	41	4	82
2	3	46	9	138
3	4	60	16	240
4	5	66	25	330
5	6	79	36	474
6	7	83	49	581
Total	27	375	139	1845

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KAIZEN MATHS

Exit ticket

Correlation and Regression

Prediction & Interpretation

Name _____

EA1

For revision time and test score, the regression line is $\hat{y} = 33.000 + 6.714x$. The observed x values were between 1 and 6. Estimate test score when revision time is 9 hours, and comment on reliability.

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KAIZEN MATHS

Exit ticket

Correlation and Regression

Prediction & Interpretation

Name _____

EB1

The table shows temperature (degrees C) and hot drinks sold (cups) for 6 observations. The regression line of y on x is $\hat{y} = 131.771 - 4.114x$. For the observation where x = 17, calculate the residual.

Observation	x	y
1	8	101
2	11	84
3	14	75
4	17	59
5	20	52
6	23	37

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KAIZEN MATHS

Exit ticket

Correlation and Regression

Prediction & Interpretation

Name _____

EC1

The table shows practice attempts (attempts) and completion time (seconds) for 6 observations. Find the regression line of y on x, then estimate completion time when practice attempts is 4 attempts. State whether this is interpolation or extrapolation.

Observation	x	y	x^2	xy
1	1	92	1	92
2	2	84	4	168
3	3	81	9	243
4	4	73	16	292
5	5	70	25	350
6	6	62	36	372
Total	21	462	91	1517

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KAIZEN MATHS

Exit ticket

Correlation and Regression

Prediction & Interpretation

Name _____

EA1

For revision time and test score, the regression line is $\hat{y} = 33.000 + 6.714x$. The observed x values were between 1 and 6. Estimate test score when revision time is 9 hours, and comment on reliability.

Show your working

Teacher answers & working

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SA1

The table shows revision time (hours) and test score (marks) for 6 observations. Find the regression line of y on x, then estimate test score when revision time is 4 hours. State whether this is interpolation or extrapolation.

Observation	x	y	x^2	xy
1	1	39	1	39
2	2	48	4	96
3	3	52	9	156
4	4	61	16	244
5	5	65	25	325
6	6	74	36	444
Total	21	339	91	1304

Answer:

$\hat{y} = 59.86$

Working:

Find the summary values from the table.

$\Sigma x = 21, \Sigma y = 339, \Sigma x^2 = 91, \Sigma xy = 1304, n = 6$

Calculate $\bar{x}$, $\bar{y}$, Sxx and Sxy.

$\bar{x} = \frac{21}{6} = 3.500$

$\bar{y} = \frac{339}{6} = 56.500$

$Sxx = \Sigma x^2 - \frac{(\Sigma x)^2}{n} = 91 - \frac{21^2}{6} = 17.500$

$Sxy = \Sigma xy - \frac{(\Sigma x)(\Sigma y)}{n} = 1304 - \frac{21 \times 339}{6} = 117.500$

Find the gradient and intercept of the regression line of y on x.

$b = \frac{Sxy}{Sxx} = \frac{117.500}{17.500} = 6.714$

$a = \bar{y} - (b \times \bar{x}) = 56.500 - (6.714 \times 3.500) = 33.000$

$\hat{y} = 33.000 + 6.714x$

Substitute the given x value into the regression line.

$\hat{y} = 33.000 + 6.714 \times 4 = 59.86$

Compare the given x value with the observed range.

Observed range: $1 \le x \le 6$. Since 4 is inside this range, this is interpolation.

Estimated test score is 59.86 marks.

Teacher answers & working

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SB1

For advertising spend and weekly sales, the regression line is ŷ = 22.000 + 9.000x. The observed x values were between 2 and 7. Estimate weekly sales when advertising spend is 10 hundreds of pounds, and comment on reliability.

Answer:

$\hat{y} = 112.00$

Working:

Substitute the given value into the regression line.

$\hat{y} = 22.000 + 9.000 \times 10 = 112.00$

Check the observed range of x.

Observed range: 2 ≤ x ≤ 7. The value 10 is outside this range.

This is extrapolation, so the prediction may be unreliable.

SC1

The table shows age of car (years) and resale value (thousands of pounds) for 6 observations. The regression line of y on x is ŷ = 24.267 - 2.171x. For the observation where x = 3, calculate the residual.

Observation	x	y
1	1	23
2	2	19
3	3	18
4	4	14
5	5	15
6	6	11

Answer:

Residual = 0.25

Working:

Use the regression line to calculate the predicted value.

$\hat{y} = 24.267 - 2.171 \times 3 = 17.75$

Use residual = observed value - predicted value.

Observed y = 18, so residual = 18 - 17.75 = 0.25

The residual is 0.25 thousands of pounds.

Teacher answers & working

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SD1

The table shows advertising spend (hundreds of pounds) and weekly sales (items) for 6 observations. Find the regression line of y on x, then estimate weekly sales when advertising spend is 5 hundreds of pounds. State whether this is interpolation or extrapolation.

Observation	x	y	x^2	xy
1	2	41	4	82
2	3	46	9	138
3	4	60	16	240
4	5	66	25	330
5	6	79	36	474
6	7	83	49	581
Total	27	375	139	1845

Answer:

$\hat{y} = 67.00$

Working:

Find the summary values from the table.

$\Sigma x = 27, \Sigma y = 375, \Sigma x^2 = 139, \Sigma xy = 1845, n = 6$

Calculate $\bar{x}$, $\bar{y}$, Sxx and Sxy.

$\bar{x} = \frac{27}{6} = 4.500$

$\bar{y} = \frac{375}{6} = 62.500$

$Sxx = \Sigma x^2 - \frac{(\Sigma x)^2}{n} = 139 - \frac{27^2}{6} = 17.500$

$Sxy = \Sigma xy - \frac{(\Sigma x)(\Sigma y)}{n} = 1845 - \frac{27 \times 375}{6} = 157.500$

Find the gradient and intercept of the regression line of y on x.

$b = \frac{Sxy}{Sxx} = \frac{157.500}{17.500} = 9.000$

$a = \bar{y} - (b \times \bar{x}) = 62.500 - (9.000 \times 4.500) = 22.000$

$\hat{y} = 22.000 + 9.000x$

Substitute the given x value into the regression line.

$\hat{y} = 22.000 + 9.000 \times 5 = 67.00$

Compare the given x value with the observed range.

Observed range: $2 \le x \le 7$. Since 5 is inside this range, this is interpolation.

Estimated weekly sales is 67.00 items.

Teacher answers & working

Correlation and Regression · Prediction & Interpretation

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EA1

For revision time and test score, the regression line is ŷ = 33.000 + 6.714x. The observed x values were between 1 and 6. Estimate test score when revision time is 9 hours, and comment on reliability.

Answer:

$\hat{y} = 93.43$

Working:

Substitute the given value into the regression line.

$\hat{y} = 33.000 + 6.714 \times 9 = 93.43$

Check the observed range of x.

Observed range: 1 ≤ x ≤ 6. The value 9 is outside this range.

This is extrapolation, so the prediction may be unreliable.

EB1

The table shows temperature (degrees C) and hot drinks sold (cups) for 6 observations. The regression line of y on x is ŷ = 131.771 - 4.114x. For the observation where x = 17, calculate the residual.

Observation	x	y
1	8	101
2	11	84
3	14	75
4	17	59
5	20	52
6	23	37

Answer:

Residual = -2.83

Working:

Use the regression line to calculate the predicted value.

$\hat{y} = 131.771 - 4.114 \times 17 = 61.83$

Use residual = observed value - predicted value.

Observed y = 59, so residual = 59 - 61.83 = -2.83

The residual is -2.83 cups.

Teacher answers & working

Correlation and Regression · Prediction & Interpretation

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EC1

The table shows practice attempts (attempts) and completion time (seconds) for 6 observations. Find the regression line of y on x, then estimate completion time when practice attempts is 4 attempts. State whether this is interpolation or extrapolation.

Observation	x	y	x^2	xy
1	1	92	1	92
2	2	84	4	168
3	3	81	9	243
4	4	73	16	292
5	5	70	25	350
6	6	62	36	372
Total	21	462	91	1517

Answer:

$\hat{y} = 74.14$

Working:

Find the summary values from the table.

$\Sigma x = 21, \Sigma y = 462, \Sigma x^2 = 91, \Sigma xy = 1517, n = 6$

Calculate $\bar{x}$, $\bar{y}$, Sxx and Sxy.

$\bar{x} = \frac{21}{6} = 3.500$

$\bar{y} = \frac{462}{6} = 77.000$

$Sxx = \Sigma x^2 - \frac{(\Sigma x)^2}{n} = 91 - \frac{21^2}{6} = 17.500$

$Sxy = \Sigma xy - \frac{(\Sigma x)(\Sigma y)}{n} = 1517 - \frac{21 \times 462}{6} = -100.000$

Find the gradient and intercept of the regression line of y on x.

$b = \frac{Sxy}{Sxx} = \frac{-100.000}{17.500} = -5.714$

$a = \bar{y} - (b \times \bar{x}) = 77.000 - (-5.714 \times 3.500) = 97.000$

$\hat{y} = 97.000 - 5.714x$

Substitute the given x value into the regression line.

$\hat{y} = 97.000 - 5.714 \times 4 = 74.14$

Compare the given x value with the observed range.

Observed range: $1 \le x \le 6$. Since 4 is inside this range, this is interpolation.

Estimated completion time is 74.14 seconds.